

Vanna-Volga FX Pricing

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1 Motivation

Black & Scholes provides adequate & fast pricing for vanilla or even barrier options, but they are different from market quotes, as the underlying assumptions don't take into volatility and rate stochasticity. This modification necessitates a more flexible modelling and calibration framework, taking into account spot-vol and smile dynamics.

More Realistic models assume stochastic rates or volatility, but this is generally hard to calibrate computationally and demands a lot of market data to find suitable parameters.

Since traders need to quote prices in a fast paced environment, they have come up with a practical rule of thumb to calibrate to market quotes, and at least avoid arbitrage to a certain degree. The idea is to quantify skew & wing vs belly backbone ATM vol dynamics using straddles, Risk reversals and Butterfly spreads. This method is fast, and produces reasonable estimates of market prices for vanillas as well as first generation exotics. It's based on adding an analytically derived correction to the B&S price, by constructing a replication portfolio, that neutralizes greeks : Vega, Vanna and Volga (thus the name) producing smile consistent values.

2 Assumptions & Replicating portfolio construction

2.1 Assumptions :

- Pricing a European Style Option O maturing in T .
- Underlying asset S_t is FX rate FOR/DOM, such that 1 unit of FOR Currency is FOR/DOM units of DOM currency. ie 1.16 EURUSD : 1 euro = 1.16 USD.
- Lognormal Dynamics : $dS_t = (r_d - r_f)S_t dt + \sigma_t S_t dW_t$
- Stochastic but strike-independent Implied Volatility

2.2 Replicating Portfolio :

Our portfolio consists of a **long** position in the option O , **short** Δ_t of the underlying and **short** x_i in 3 pivot European options (ATM Straddle - RR_{25d} Risk reversals and BF_{25d} Butterfly Spreads. The change in the replicating Portfolio is :

$$d\Pi_t = dO_t - \Delta_t dS_t - \sum_{i=1}^3 x_i dC_i$$

Applying Itô's Lemma :

$$\begin{aligned} d\Pi = & \left[\frac{\partial O}{\partial S_t} - \Delta_t - \sum_{i=1}^3 x_i \frac{\partial C_i}{\partial S_t} \right] dS_t \\ & + \left[\left(\frac{\partial O}{\partial t} - \sum_{i=1}^3 x_i \frac{\partial C_i}{\partial t} \right) + \frac{1}{2} \sigma_t^2 S_t^2 \left(\frac{\partial^2 O}{\partial S_t^2} - \sum_{i=1}^3 x_i \frac{\partial^2 C_i}{\partial S_t^2} \right) \right] dt \\ & + \left[\frac{\partial O}{\partial \sigma_t} - \sum_{i=1}^3 x_i \frac{\partial C_i}{\partial \sigma_t} \right] d\sigma_t + \frac{1}{2} \left[\frac{\partial^2 O}{\partial \sigma_t^2} - \sum_{i=1}^3 x_i \frac{\partial^2 C_i}{\partial \sigma_t^2} \right] (d\sigma_t)^2 \\ & + \left[\frac{\partial^2 O}{\partial S_t \partial \sigma_t} - \sum_{i=1}^3 x_i \frac{\partial^2 C_i}{\partial S_t \partial \sigma_t} \right] dS_t d\sigma_t. \end{aligned}$$

By no arbitrage this portfolio is locally risk free ie :

$$d\Pi_t = r_d \Pi_t dt$$

And we neutralize portfolio sensitivities to Delta, Vega, Vanna and Volga. This gives the following system :

$$\Delta_t = \left[\frac{\partial O}{\partial S_t} - \sum_{i=1}^3 x_i \frac{\partial C_i}{\partial S_t} \right]$$

$$\mathbf{w}_O = \mathbf{V} \mathbf{x}$$

where

$$\mathbf{w}_O = \begin{pmatrix} \text{Vega}(O) \\ \text{Vanna}(O) \\ \text{Volga}(O) \end{pmatrix} = \begin{pmatrix} \frac{\partial O^{BS}}{\partial \sigma_t} \\ \frac{\partial^2 O^{BS}}{\partial S_t \partial \sigma_t} \\ \frac{\partial^2 O^{BS}}{\partial \sigma_t^2} \end{pmatrix},$$

$$\mathbf{V} = \begin{pmatrix} \text{Vega}_1 & \text{Vega}_2 & \text{Vega}_3 \\ \text{Vanna}_1 & \text{Vanna}_2 & \text{Vanna}_3 \\ \text{Volga}_1 & \text{Volga}_2 & \text{Volga}_3 \end{pmatrix} = \begin{pmatrix} \frac{\partial C_1^{BS}}{\partial \sigma_t} & \frac{\partial C_2^{BS}}{\partial \sigma_t} & \frac{\partial C_3^{BS}}{\partial \sigma_t} \\ \frac{\partial^2 C_1^{BS}}{\partial S_t \partial \sigma_t} & \frac{\partial^2 C_2^{BS}}{\partial S_t \partial \sigma_t} & \frac{\partial^2 C_3^{BS}}{\partial S_t \partial \sigma_t} \\ \frac{\partial^2 C_1^{BS}}{\partial \sigma_t^2} & \frac{\partial^2 C_2^{BS}}{\partial \sigma_t^2} & \frac{\partial^2 C_3^{BS}}{\partial \sigma_t^2} \end{pmatrix}.$$

The strike K of the option O can be different from the strikes K_i ; the Maturities T_i as well can be different; but this introduces a certain liberty/axing in dynamics; as back end T is more sensitive to Gamma while long end T is more sensitive to Vega.

It is important to note that the assumption of stochastic but strike independent volatility is what allows this modelling framework, in fact a proof is given by Shkolnikov 2009 based on the equality of the portfolios at maturity, a backward induction based on locally risk free portfolios (consistency argument) at previous timesteps proves that the portfolios agree on each time point t_j .

2.3 FX Market Instruments :

To simplify notation, we denote $C(K), P(K)$ the call and put of strike K , maturity T, and define :

- ATM straddle : $C_1 = C + P$
- 25d Risk Reversal : $C_2 = C_{K_1} - P_{K_2}$ with K_1 and K_2 chosen so that the options have a delta of .25 in absolute value.
- 25d Butterfly $C_3 = \beta[(C_{K_1} + P_{K_2}) - (P(S) - C(S))]$ defined as with determined to set the vega of the butterfly to 0.

This system is popular because the benchmark securities are very liquid, and because the resulting matrix of is almost diagonal, which allows an intuitive interpretation of the coefficients x_i .

2.4 Practicalities & Notes

- FX is a particular asset class, Central Banks frequently intervene usually in the spot market to protect pegs and manage their currencies. The bank's goal is to keep the exchange rate in a within a certain band, the resources the bank spends end up as the P&L of some other market participant whose goal is to generate profit. This asymmetry is key for the importance of the risk reversal.